

Synergizing IoT, AI, and blockchain for smart agriculture: Challenges, opportunities, and future directions

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ABSTRACT

The integration of the Internet of Things (IoT), blockchain technology (BT), and Artificial Intelligence (AI) is transforming agriculture into a smart, data-driven system designed to enhance productivity, transparency, and automation. Population growth and limited resources make these technologies increasingly critical, especially in regions with scarce water, nutrients, or fertile soil. IoT provides real-time monitoring and physical data collection through sensors and edge devices, BT ensures data security, traceability, and transparency across supply chains, while AI enables predictive analytics and automated decision-making, reducing direct farmer intervention. This systematic literature review is focusing on the IoT implementations in the agriculture ecosystem, with the sole aim of increasing agricultural productivity and efficiency. Furthermore, it analyzes the interplay of IoT, AI, and BT in agriculture, with the emphasis on the measurable impacts, security of communication protocols, socio-technical implications, and automation and decision-making, among others. Despite their promise, integration faces notable barriers such as data privacy, interoperability, real-time processing, and implementation costs. Using the PRISMA framework, 35 studies were selected from an initial pool of 977 articles published between 2019 and 2025. A rigorous quality assessment extracted insights on integration strategies, technical limitations, and practical applications. The review highlights opportunities and challenges in adopting IoT, AI, and BT for sustainable smart agriculture. It concludes with recommendations for researchers, policymakers, technology developers, and practitioners to address current gaps, strengthen security and interoperability, and guide future advancements toward resilient and efficient agricultural systems.

1. Introduction

The global agricultural sector faces unprecedented challenges, including ensuring food security for a growing population, mitigating the impacts of climate change, and optimizing resource utilization [1]. Traditional farming methods often lead to negative consequences such as decreased soil fertility and potential harm to human health due to uncontrolled use of pesticides and water. To address these complex issues, the agricultural landscape is undergoing a profound transformation driven by the principles of Industry 4.0 and advanced digital technologies [2]. The integration of cutting-edge technologies like the Internet of Things (IoT), blockchain technology (BT), and Artificial Intelligence (AI) is pivotal in creating more productive, secure, and automated agricultural ecosystems.

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IoT facilitates real-time monitoring and data collection through various sensors and robots, enabling remote management of agricultural processes [3].

The evolution of agriculture reflects this technological progression, from primitive tools in Agriculture 1.0 to the advanced machinery of Agriculture 2.0, and the intelligent systems of Agriculture 3.0, which introduced computer programs, automated irrigation, and GNSS (Global Navigation Satellite System). Agriculture 4.0 marks a significant shift towards digitalization and intelligent systems, leveraging technologies such as Blockchain, IoT, and AI for remote monitoring and automation. Looking ahead, Agriculture 5.0 envisions smart agriculture deeply rooted in automation, four-dimensional communication, and long-term strategies powered by advanced robotics, AI, and 6 G networks [4,5].

The impact of climate change on agriculture is profound, influencing both the reduction and increase of food production. Conversely, agricultural practices themselves contribute to climate change, particularly through the emission of gases like methane and nitrous oxide from livestock and fertile land management. It is also worth noting that one of the climate problems is the change in sea level, which often causes fertile land to be inundated, and salt water penetrates even into dry areas where there is a lack of fresh water, which also poses a threat to the production of agricultural products [6]. Agriculture is one of the industries that most utilizes water resources, and one of the primary materials for cultivating agricultural products [7].

Furthermore, agricultural expansion often necessitates deforestation, leading to a reduction in biodiversity. Therefore, developing long-term strategies using contemporary technology is crucial not only for increasing productivity but also for ensuring the safety of agricultural products, thereby fostering a healthier global ecosystem [8].

The development of agriculture is also vital for national survival and food security, playing a significant role in economic growth by contributing to gross domestic product. Given the projected global population increase to 10 billion by 2050, as estimated by the United Nations, a lack of long-term strategies for enhancing agricultural product, water, and nutrient productivity could lead to severe human survival challenges [9]. Consequently, major global powers are investing heavily in smart agricultural technologies and encouraging youth participation in the sector. Numerous smart agriculture applications are already in use today, including intelligent irrigation, water management, soil management, climate management, crop management, and nutrient management, all contributing to increased agricultural productivity [10].

Smart Agriculture (SA) aims to revolutionize agricultural practices, because it tends to go beyond the optimization of existing techniques. Recent applications of AI enable real monitoring, where through modern analysis and techniques, the most ideal, sustainable, healthiest planting conditions can be predicted. Above all, AI contributes to reducing costs, reducing waste, and increasing productivity [11]. Despite the promising advancements and numerous implementations of IoT devices in smart agriculture, several open issues and challenges persist. These include concerns related to energy consumption and the adoption of efficient methods for rural areas, the significant cost of purchasing and maintaining intelligent devices, and the complexities of managing high-capacity data in real-time. Furthermore, challenges arise from diverse communication protocols, data privacy and security, ethical considerations, scalability issues, data access limitations, and the notable absence of global standards. The effective integration of Blockchain and AI with existing systems also presents considerable hurdles [12,13].

This systematic literature review aims to comprehensively analyze and synthesize the current state of research on the integration of IoT, BT, and AI for developing secure and automated agricultural systems. By systematically reviewing existing literature, we seek to identify current applications, prevalent challenges, proposed solutions, and future research directions in this interdisciplinary field.

To achieve this aim, the following research questions will guide our investigation:

- **RQ1:** What are the available publication channels, including high-impact factor journals, that disseminate research on the integration of IoT, blockchain and AI in smart agriculture settings, and what are the recent publication trends in the field?
- **RQ2:** How have IoT technologies been practically implemented in smart agriculture, and what measurable impacts (e.g., productivity and cost-effectiveness) have they demonstrated?
- **RQ3:** What are the prevailing communication protocols between IoT devices and physical agricultural systems, and what are the associated security vulnerabilities and mitigation strategies?
- **RQ4:** What solutions or frameworks have been proposed to address the security, integrity, and automation arising from the synergistic integration of IoT, Blockchain and AI technologies in agriculture?
- **RQ5:** What are the socio-technical barriers that impede a successful deployment of IoT, blockchain and AI in smart agriculture?

The remainder of this paper is structured as follows: Section 2 highlights the state-of-the-art of IoT, blockchain and AI integration in agriculture, whereas Section 3 details the methodology employed for this systematic literature review. Section 4 presents the research question's findings, discusses the identified challenges and proposed solutions, along with some recommendations and future research directions. Finally, Section 5 concludes the review and provides a summary of the key findings.

2. Background and related work

The agricultural sector faces increasingly complex challenges, driven by factors such as global population growth, the conversion of fertile land for urban development, and concerns related to sustainable resource management [14]. These pressures necessitate a fundamental transformation in farming practices, with a growing reliance on advanced digital solutions. Agronomists, for instance, are increasingly focusing on developing innovative techniques, such as cultivating products in simulated conditions within smart greenhouses or indoor farming environments [15,16], which inherently demand sophisticated technological integration for optimal control and productivity. This shift necessitates a comprehensive understanding of key enabling technologies: the IoT for ubiquitous

data collection, AI for intelligent analysis and automation, and blockchain technology for ensuring data integrity, trust, and traceability [4,5].

In the subsequent subsections, each of these technologies, along with the imperative for their synergistic integration, will be presented in detail, laying the groundwork for understanding their combined role in secure and automated agricultural systems.

2.1. Internet of things (IoT) in agriculture

IoT is a rapidly expanding field finding significant applications across various sectors, including agriculture, due to its capability to enable real-time data collection and remote management. IoT devices, comprising intelligent sensors, robots, and Unmanned Aerial Vehicles (UAVs), facilitate real-time monitoring of agricultural products and precise measurement of various soil characteristics, water content, dryness, and other vital nutrients.

Key applications of IoT in smart agriculture include *precision farming*, which involves monitoring soil moisture and nutrient levels for optimized irrigation and resource management [17]. Another crucial application is *crop monitoring*, focusing on detecting pests and diseases, tracking crop growth, and sensing environmental factors such as temperature and humidity [18]. *Livestock management* benefits from IoT through automatic milking, automated feed systems, animal tracking, disease monitoring, and behavioral prediction [19]. Furthermore, *automated machinery* utilizes Unmanned Ground Vehicles (UGVs) and other robots for tasks like spraying and image capture, though commanding them over vast lands presents challenges [20]. Lastly, *smart greenhouses* and Indoor Farming leverage IoT through techniques such as hydroponics, aeroponics, and aquaponics to cultivate products in controlled environments, thereby increasing productivity [21].

2.2. AI in agriculture

Artificial intelligence (AI) serves as a critical component in the digitalization of agriculture, primarily focused on automating every aspect of farm management and creating long-term strategies through advanced data analysis, forecasting, and real-time monitoring. AI, particularly through its subfields like machine learning and deep learning (DL), helps to reduce the reliance on pesticides, optimize the use of water and other resources, and improvises various models for predictive analysis.

Specific applications of AI are revolutionizing various aspects of farming. One key area is *predictive analytics*, which enables the forecasting of crop yields, the prediction of rainfall patterns, and the identification of necessary nutrients for optimal growth [22]. This data-driven approach helps farmers make informed decisions and manage risks effectively. *Disease and pest detection* is another critical application, allowing for the early detection of threats to agricultural products through advanced image processing and other analytical methods [23]. AI-powered systems can identify signs of disease or pests in real-time, enabling prompt intervention and reducing crop losses. Furthermore, AI contributes significantly to *resource optimization*, facilitating intelligent irrigation scheduling, precise fertilizer recommendations, and comprehensive nutrient management. This ensures that resources like water and fertilizers are applied efficiently, minimizing waste and promoting sustainable practices. *Genomic and phenotypic analysis* leverages advanced AI analysis of products to improve crop improvisation and reduce environmental impact. Lastly, *automation* involves driving the autonomous actions of intelligent machines, with the ultimate goal of achieving complete automation of agricultural processes [24]. This includes tasks like automated harvesting, weeding, and even milking in livestock operations.

The synergy between AI and IoT, often referred to as the Artificial Intelligence of Things (AIoT), has found significant application in smart agriculture, particularly for assessing crop yields and detecting diseases early. AIoT aims to automate the decisions made by intelligent devices based on data collected from IoT sensors [18].

2.3. Blockchain technology in agriculture

Blockchain technology is increasingly recognized as a foundational element for ensuring data protection, privacy, and automation in agricultural systems through its decentralized and cryptographic mechanisms. BT strengthens the agricultural food chain by enabling automation of contracts, eliminating intermediaries, and significantly increasing transparency, reliability, traceability, and data immutability [25]. A core application is *supply chain traceability and authenticity*, which provides a secure and transparent chain to track agricultural products from their origin, including cultivation methods, nutrients used, and certifications, until they reach the consumer [26]. This enhanced traceability helps combat fraud and ensures product authenticity. Furthermore, blockchain technology offers robust *data security and integrity* due to its decentralized nature and cryptographic mechanisms, where data is distributed across all nodes, thus offering strong protection against cyberattacks [27].

Blockchain also facilitates *automated services* by enabling the automation of processes through smart contracts without the need for third-party intervention, although practical implementation still faces challenges. Finally, the technology fosters *trust and transparency* among participants in the agricultural chain, even those who are otherwise unknown to each other, through distributed ledgers and transparent transaction verification. Despite its immense potential, the practical implementation of blockchain in agriculture faces hurdles related to the complexity of smart contracts, scalability issues, and the need for standardized frameworks for sustainable practices [28].

2.4. Synergy and integration of IoT, blockchain and AI in agriculture

The seamless integration of IoT, AI, and blockchain in agriculture is now considered inevitable, as these technologies collectively

address the limitations faced by each when implemented in isolation. The synergy aims to create truly secure, transparent, automated, and productive agricultural systems, as following:

- **IoT as the data foundation:** IoT devices generate vast amounts of real-time data on plant health, soil characteristics, livestock, and environmental conditions.
- **AI for intelligent decision-making and automation:** AI processes and analyzes the secure and authenticated data from IoT and blockchain to provide predictive insights, optimize resource allocation, automate tasks, and enable intelligent machines to make autonomous decisions. AI can also optimize blockchain network performance and analyze complex transactional data for market insights.
- **Blockchain for data security and trust:** Blockchain technology secures and authenticates this high-capacity data, ensuring its integrity, privacy, and traceability throughout the agricultural chain. It provides a tamper-proof record of all activities, from cultivation to distribution, fostering transparency and preventing fraud.

Fig. 1 illustrates the multifaceted synergy between IoT, AI, and blockchain, highlighting their combined contribution to increased productivity, security, and automation in agriculture. The figure, in addition to illustrating the implementations of each technology separately, also illustrates the benefits from the synergy of these three technologies in agriculture, which contributes above all to increasing efficiency, safety, and sustainability through intelligent, safe, and transparent actions.

The journey from the need for automated and secure agriculture to its full implementation involves several key activities, including identifying current agricultural challenges, determining tasks for automation and security, evaluating infrastructure, selecting appropriate technologies, applying security measures, and continuous performance monitoring. Despite the clear benefits, significant barriers to adoption persist, such as the digital divide, high upfront costs, complexity of integration, and ongoing privacy and security concerns [29,30].

2.5. Related literature reviews

The practical implementation of smart agriculture, particularly concerning the synergy between IoT, AI, and blockchain is progressing, but it faces numerous practical challenges, especially related to initial integration. The body of knowledge has evidenced various empirical analyses and systematic literature reviews that have been conducted to explore different aspects of this field. Table 1 provides a summary of relevant past reviews and their primary focus areas. In addition to the synergy and implementation of IoT, AI and BT, a particular emphasis and importance during the analysis and comparison of literatures is also given to the communication mechanisms, security and mitigation strategies, and above all productivity and cost-effectiveness.

As can be seen, existing reviews have provided valuable insights into specific facets of smart agriculture. However, a comprehensive review that simultaneously addresses productivity, cost-effectiveness, communication protocols, security vulnerabilities and mitigation strategies, the role of AI in smart contracts and automation, blockchain for data security and integrity, has been largely

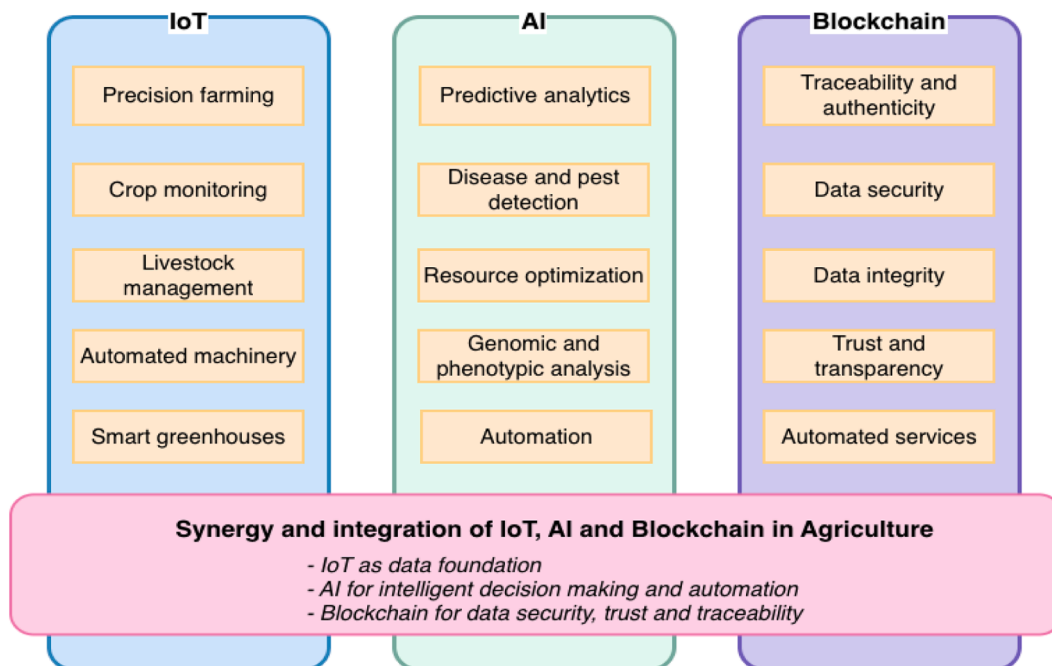


Fig. 1. IoT, AI, and blockchain integration in agriculture settings.

Table 1

Summary of retrieved past related reviews.

Surveys	IoT implementation	Measurable impacts (productivity and cost-effectiveness)	Communication protocols between IoT and agricultural systems	Security vulnerabilities and mitigation strategies	AI for smart contract and automation	Blockchain for data security and integrity	AI and Blockchain integration
[1]	Yes	productivity	No	No	No	Yes	No
[4]	Yes	cost-effectiveness	Yes	Yes	Yes	No	No
[5]	Yes	productivity	No	Yes	No	Yes	No
[10]	Yes	No	No	No	Yes	No	No
[12]	Yes	No	Yes	Yes	Yes	Yes	No
[15]	Yes	No	Yes	Yes	Yes	Yes	No
[16]	Yes	No	No	No	No	No	No
[17]	Yes	Yes	No	No	Yes	No	No
[18]	Yes	No	Yes	No	Yes	No	No
[20]	Yes	productivity	No	Yes	No	Yes	No
[22]	Yes	productivity	Yes	No	Yes	No	No
[24]	Yes	No	No	No	Yes	Yes	No
[27]	Yes	No	Yes	Yes	Yes	Yes	No
[31]	Yes	No	Yes	Yes	Yes	No	No
[32]	Yes	No	No	No	No	No	No
[33]	Yes	cost-effectiveness	Yes	Yes	Yes	No	No
[34]	Yes	No	Yes	Yes	No	No	No
[35]	Yes	No	Yes	No	Yes	No	No
[36]	Yes	No	Yes	Yes	No	Yes	No
[37]	Yes	No	No	Yes	No	No	No
[38]	Yes	No	Yes	No	No	No	No
[39]	Yes	No	Yes	Yes	No	No	No
[40]	Yes	No	Yes	Yes	Yes	No	No
[41]	Yes	No	Yes	No	No	No	No
[42]	Yes	No	Yes	Yes	Yes	No	No
[43]	Yes	No	Yes	Yes	Yes	No	No
[44]	Yes	No	Yes	Yes	Yes	No	No
[45]	Yes	No	No	Yes	Yes	No	No
[46]	Yes	No	Yes	No	No	No	No
[47]	Yes	productivity	No	Yes	No	Yes	No
[48]	Yes	No	No	Yes	No	Yes	No
[49]	Yes	productivity	No	Yes	No	No	No
[50]	Yes	No	Yes	Yes	Yes	No	No
[51]	Yes	No	Yes	Yes	No	Yes	No
[52]	Yes	cost-effectiveness	No	Yes	No	No	No
[53]	Yes	Yes	Yes	No	Yes	Yes	No
[54]	Yes	No	Yes	Yes	No	Yes	No
[55]	Yes	productivity	Yes	No	Yes	No	No
[56]	Yes	Yes	Yes	Yes	Yes	No	No
[57]	Yes	productivity	Yes	Yes	Yes	Yes	No
[58]	Yes	cost-effectiveness	No	No	Yes	Yes	No
[59]	Yes	No	No	Yes	No	No	No
[60]	Yes	No	Yes	No	No	No	No
[61]	Yes	No	No	No	Yes	No	No
[62]	Yes	No	Yes	Yes	No	Yes	No
[63]	Yes	productivity	Yes	Yes	No	Yes	No
[64]	Yes	Yes	Yes	Yes	No	No	No
[65]	Yes	No	No	Yes	No	Yes	No
Our work	Yes	Yes	Yes	Yes	Yes	Yes	Yes

missing. Critically, the holistic integration of IoT, AI, and blockchain has been largely missing. Our work distinguishes itself by offering such integrated perspective, aiming to provide a detailed understanding of the current status, practical challenges, and future improvements for the complete automation and security of agricultural systems.

3. Methodology

This systematic review analyses the implications of IoT, AI, and blockchain for secure and automated agricultural systems. Relevant studies were systematically identified from selected repositories to meet the research objectives and ensure a high standard of review quality. Additionally, the study adheres to the guidelines outlined by [66], encompassing the subsequent steps:

1. Search Strategy
2. Identification of Primary Studies
3. Information Extraction - Screening
4. Threats to Validity

3.1. Developing search strategy

To identify relevant literature on the subject, we examined two prominent research repositories: "Scopus," and the "Web of Science." These repositories include most research publications related to the five specified research questions. Considering the review scope, we construct the search query using three categories of terms outlined in Table 2.

Additionally, each group was conjoined using the "AND" operator. The search terms across the three categories were amalgamated using the "OR" operator. "Group A" emphasizes the Internet of Things. Conversely, "Group B" emphasizes smart agriculture. "Group C" encompasses terminology associated with communication protocols, blockchain and AI. We used uniform search keywords across all two databases.

3.2. Identification of initial findings

Fig. 2 illustrates the PRISMA diagram of our investigation, depicting the whole process from the first data extraction to the filtration of pertinent data. It provides a comprehensive summary of the flow of information, including the quantity of detected records, those screened, appraised for eligibility, and ultimately included in the final analysis.

We performed literature searches using the title, abstract, and author keywords to identify relevant material. Executing the query yielded a total of 977 results across all databases, as seen in Fig. 2. A total of 772 articles were obtained from Scopus, representing the maximum quantity, and 205 articles were obtained from Web of Science. It is worth noting that we have taken as a basis only the two databases Scopus and Web of Science, considering that most papers published in conferences or various journals are indexed in Scopus among others, although the same can also be indexed in IEEEExplore, SpringerLink, ACM/DL, but Scopus is known as one of the databases that index the majority of papers. Also, when classifying papers and searching based on the search string, we are limited to peer-reviewed journal and conference papers.

3.3. Screening phase

At this point, we implemented the inclusion and exclusion criteria and finalized the studies that met the specified criteria. Initially, we deleted duplicates from the chosen research, then removed publications not authored in English, and ultimately omitted papers that fell outside our specified time window (2019–2025). Furthermore, we performed title and abstract screening, excluding articles unrelated to the topic. Additionally, we omitted survey articles and studies without full-text availability. Furthermore, we excluded the articles that did not meet the quality evaluation criteria, explained in the next section.

3.3.1. Defining quality assessment criteria

Quality Assessment Criteria (QAC) have been established to evaluate the quality of research for inclusion in the systematic

Table 2
Search query formation.

Search group	Search criterion
Group A:	("Internet of Things" OR IoT)
Group B:	("smart agriculture" OR "precision farming" OR "digital agriculture")
Group C:	((implement* AND (productiv* OR "cost effectiveness")) OR ("communication protocol*" OR "network protocol*") AND (secur* OR vulnerab*)) AND (blockchain AND (integrat* OR "data integrity" OR traceab*)) AND ((AI OR "artificial intelligence") AND (automat* OR "human replacement"))
Search query	"(Group A)" AND "(Group B)" AND "(Group C)"

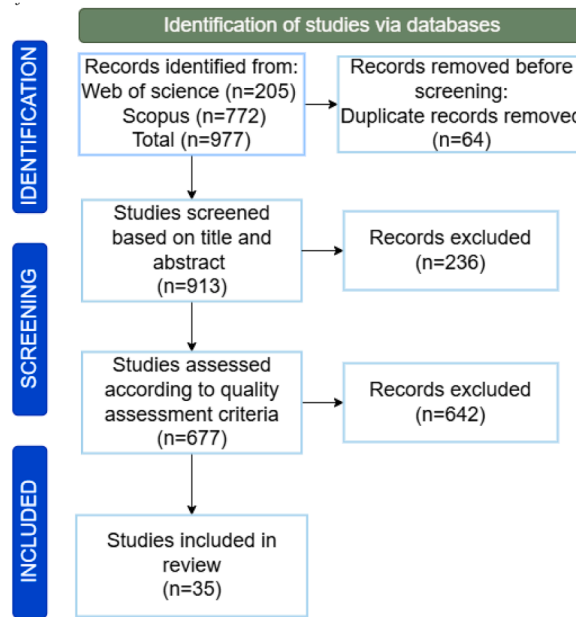


Fig. 2. PRISMA framework.

Table 3
Quality assessment criteria.

Item #	QA question	Score	Description
QAC1	Are the research objectives clearly articulated?	0 points	No, objectives are not stated.
		0.5 points	Partially, objectives are stated but not clearly.
		1 point	Yes, objectives are clearly stated.
QAC2	Is the research methodology well-defined?	0 points	No, the research methodology is not properly defined.
		0.5 points	Partially, the methodology is defined but does not mention relevant steps for methodology.
		1 point	Yes, the methodology is properly defined.
QAC3	Does the study explicitly address the specific domains of smart agriculture and the measurable impacts, such as productivity and cost-effectiveness?	0 points	No, the study does not explicitly address any domain in which smart agriculture intersects with agricultural productivity and cost-effectiveness.
		0.5 points	Partially, the study addresses the specific domains in which smart agriculture intersects with agricultural productivity and cost-effectiveness, but unclearly.
		1 point	Yes, study explicitly addresses the specific domains in which smart agriculture intersects with agricultural productivity and cost-effectiveness.
QAC4	Does the study thoroughly analyze communication protocols in agricultural IoT, and their security risks?	0 points	No, the study does not explicitly address any communication protocols and the associated risks.
		0.5 points	Partially, the study does not explicitly address any communication protocols and the associated risks, but unclearly.
		1 point	Yes, the study explicitly addresses communication protocols and the associated risks.
QAC5	Does the study properly discuss the synergetic integration of IoT, AI and blockchain in agricultural systems?	0 points	No, the study does not properly discuss the synergetic integration of IoT, blockchain and AI in agricultural systems
		0.5 points	Partially, the study properly discusses the synergetic integration of IoT, blockchain and AI in agricultural systems, but unclearly.
		1 point	Yes, the study properly discusses the integration of IoT, blockchain and AI in agricultural systems.
QAC6	Does the conclusion reflect the research findings and future research trends?	0 points	No, the conclusion does not reflect the research findings and future research trends.
		0.5 points	Partially, the conclusion reflects the research findings and future research trends, but not properly and clearly.
		1 point	Yes, the conclusion reflects the research findings and future research trends.

literature review. Each research publication is thus assessed in accordance with the specified QAC. Every QAC corresponds to a Quality Assessment (QA) question. Each QA question may be answered with *yes*, *partly*, or *no*, with corresponding numeric scores: *yes* receives a score of 1, *partly* receives 0.5, and *no* receives 0. The aggregate score of each research is determined by summing the points for responses to specified QA questions. Table 3 provides a concise overview of QAC. It is worth noting that 677 papers out of a total of 977 were part of the QAC process.

3.3.1.1. Quality assessment of studies. Primary studies were chosen after screening and eligibility assessments, namely those that satisfied the inclusion and exclusion criteria. Based on the QAC outlined in Table 3, the majority failed to satisfy the QAC due to insufficient specification of the relevant topics employed in their research. Consequently, studies that received a score below 4 based on QA questions were excluded from further data extraction. As a result, only studies that satisfy QAC and get a score of 4 or higher were considered for data extraction and synthesis. In total, 35 studies satisfied the QAC criteria and were subsequently used for data extraction and synthesis.

3.3.2. Data extraction and synthesis

Following the selection of studies that satisfy QAC, pertinent data or information related to the research questions is retrieved. Table 4 shows the data elements to be collected from publications along with their relevance to the study topics. Data items like journal name, publisher, impact factor, communication protocols, digital technologies (IoT, AI, blockchain), and socio-technical implications hindering automation are extracted from selected studies.

3.4. Reporting phase - included studies

Each research question is examined comprehensively, and a summary of the gathered data (findings) is provided in a designated section. RQ1, concerning publication channels, is addressed in Section 4.1. RQ2, regarding the IoT integration, is examined in Section 4.2. RQ3, which pertains to the communication protocols and security, is discussed in Section 4.3. RQ4, related to the synergetic application of IoT, AI and blockchain in agriculture, is covered in Section 4.4, whereas findings regarding socio-technical barriers, pertaining to RQ5 are covered in Section 4.5. Furthermore, a comprehensive analysis of the challenges, benefits, and economic and business implications is elaborated in Sections 4.6 and 4.7.

3.5. Limitations and threat to validity

Database selection: The primary research databases used for article selection in our study were Scopus and Web of Science. These databases prioritize trustworthiness and relevance to the subject of this study, ensuring access to a diverse collection of peer-reviewed, high-quality articles across several disciplines. While our focus was on these research databases, it is crucial to recognize additional repositories, such as Google Scholar, renowned for its comprehensive indexing of research articles. This database was excluded from the study. Furthermore, this study only includes peer-reviewed journal papers and conference proceedings. Scientific studies that are not peer-reviewed are not included.

Timeframe for the selected research: The timeframe for the chosen studies encompasses the previous 5 years, namely from January 1, 2020, to March 1, 2025. We omitted publications published before 2020 since our initial search yielded insufficient relevant studies that might adequately inform our research questions, facilitate a critical evaluation of the evidence, and support appropriate findings. We underline that conference papers presented before March 1, 2025, which may not have been published by the study's cut-off date, were removed from our literature review.

Language consideration: We omitted articles by concentrating just on those authored in English. Language restrictions limit our study, perhaps leading to the exclusion of significant findings written in a language other than English.

Table 4
Data extraction strategy.

Data item no.	Data item/aim	Description	Relevancy
1	Journal, publisher, and impact factor on papers with relevant topic	To find publication channels for research dealing with IoT, blockchain and AI integration in agriculture	RQ1
2	IoT integration in agriculture and measurable impacts	To find the publication for research on the topic relating to IoT integration and measurable impacts	RQ2
3	Communication protocols and security	To find the publication on communication protocols between IoT devices and physical agricultural systems, and what are the associated security vulnerabilities and mitigation strategies	RQ3
4	Synergetic application of digital technologies (IoT, blockchain and AI) in agricultural systems	To find papers examining integration of IoT, AI and blockchain in agricultural systems, and how does it enhance security, data integrity, and automation	RQ4
5	Socio-technical limitations for successful deployment of IoT, AI, and blockchain	To find publication explaining the potential socio-technical limitations	RQ5

4. Results and discussion

4.1. Available publication channels and research trends (RQ1)

This section discusses the available publication channels and research trends for integration of IoT, AI, and blockchain in agriculture. It highlights publishers, their various journals, and their impact factors as reported in the Journal Citation Report 2024, alongside with several research papers on IoT, AI and blockchain in agriculture, as seen in Table 5. A total of 35 papers were chosen; from which 34 were published in journals with an impact factor.

The number of retrieved papers across different regions/countries of the first author is presented in Fig. 3. Regions having the greatest density of research works are shown with a darker color on the map. By observing the initial author's place of origin, most of the works are from Asia, with 18 papers, followed by Europe with 7 papers, Africa with 6 papers, Australia with 3 papers, and North America with 1 paper. The majority of research in Asia originates from India. In Europe, Spain is represented with most research works published, in Africa, Morocco is the country with most published papers, whereas the USA is the country representing North America.

Fig. 4 illustrates the yearly academic publishing of papers and the evolving trend in the research domain from 2020 to 2025. From Fig. 4 we see that the year 2024 dominates with 14 papers published, followed by the year 2023 with 9 papers, the year 2022 with 5 papers, the years 2021 and 2025 with 3 papers each, and finally the year 2020 with 1 paper.

In Table 6, we present a detailed summary of the 35 included papers, categorized by objectives, and relation to the RQs.

4.2. IoT implementation and the measurable impacts (RQ2)

IoT technologies are practically revolutionizing smart agriculture by enhancing efficiency and productivity through precise data collection, automated decision-making, and optimized resource use. This transformation also includes significant improvements in cost-effectiveness and environmental sustainability. A key practical implementation involves the synergistic combination of IoT, BT, cloud technology, and AI techniques, where the transfer of data from IoT sensors to decentralized storage is facilitated by oracles and the MQTT (Message Queue Telemetry Transport) communication protocol [2]. Oracles act as a bridge, connecting cloud technology and AI for big data analytics, the blockchain network, implemented through smart contracts - decentralized applications, and various agricultural product trading companies. To bolster supply chain transparency for consumers, a proposed system suggests assigning each product a QR code; scanning it would reveal its entire cultivation history stored on the blockchainCommunication technologies, which in this case are essential for smart agriculture include Lo-RaWAN, ZigBee, and BACnet, which are favored due to their low consumption, compact size, ease of implementation, scalability, and effective communication ranges [75]. Hardware components commonly found in IoT implementations span Raspberry Pi, NodeMCU, Arduino Uno/Mega/Nano, along with various sensors like DHT11, RTC modules, GSM modules, LM35, and EC sensors [67].

Measurable impacts are evident in smart irrigation systems, where IoT-based approaches have been shown to reduce water use by

Table 5
Available publication channels.

Publisher	Journal name	Impact factor	# of studies
Elsevier	Internet of Things	7.6	1
	Information & Communications Technology Express	4.2	1
	Computers and Electronics in Agriculture	8.9	2
	Computers & Electrical Engineering	4.9	1
	Results in Engineering	7.9	2
	Scientific African	3.3	1
	Journal of Information Security and Applications	3.7	1
	Sustainable computing	5.7	1
	Blockchain: Research and Applications	5.6	1
	Alexandria Engineering Journal	6.8	1
IEEE	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	5.3	1
	IEEE Access	3.4	7
	IEEE Open Journal of the Computer Society	8.2	1
	IEEE Internet of Things	8.9	1
Springer	Operations Management Research	5.3	1
	Journal of King Saud University - Computer and Information Sciences	6.1	1
	SN Applied Sciences	2.8	1
	Wireless Personal Communications	2.2	1
Wiley Online Library	Wireless Communications and Mobile Computing	2.146	1
	Journal of Sensors	1.1	1
	Computational Intelligence and Neuroscience	2.284	1
	Internet Technology Letters	0.5	1
	IET Collaborative Intelligent Manufacturing	3.1	1
	Sensors	3.5	2
MDPI	Sustainability	3.3	1
Editorial Salud, Ciencia y Tecnología	Data and Metadata	–	1
Total studies			35

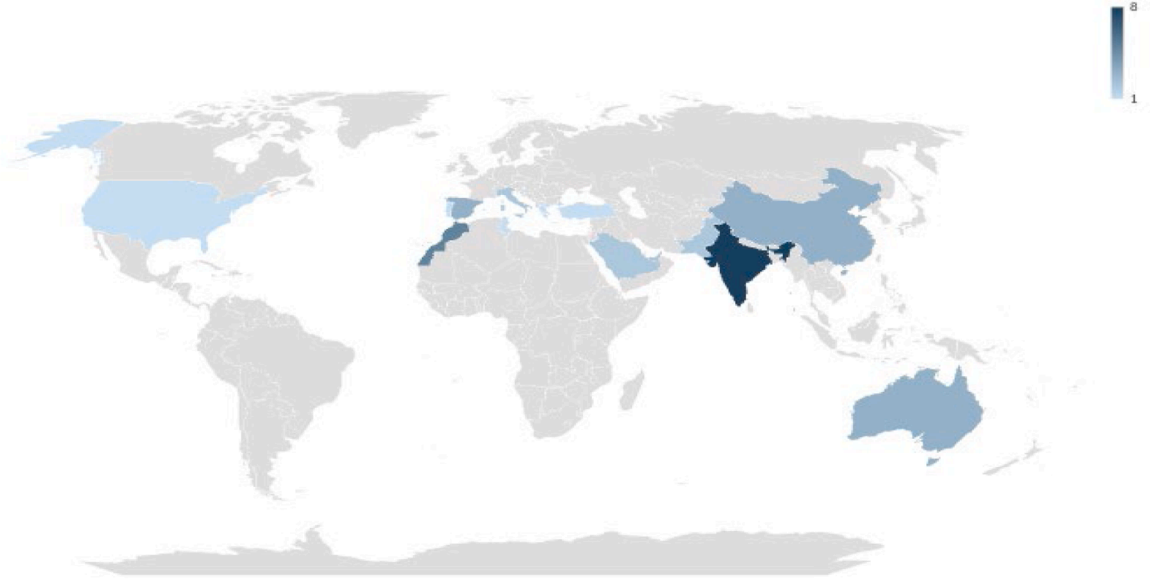


Fig. 3. Geographical distribution of papers that are part of the study.

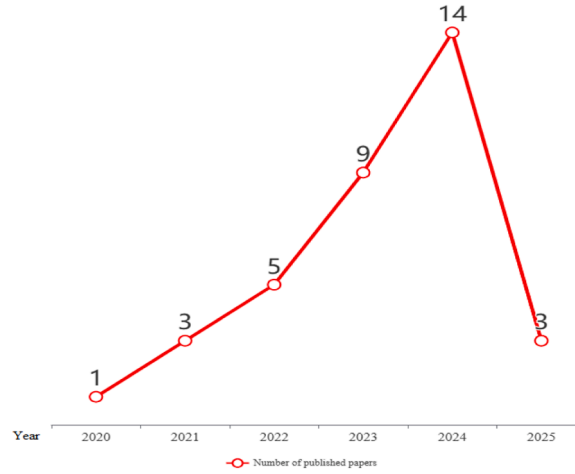


Fig. 4. Yearly academic publications of papers.

up to 50 % without compromising yields. For instance, a smart irrigation framework for sensitive crops like tomatoes utilizes the Message Queuing Telemetry Transport system for data transmission and proposes a Reinforcement Learning (RL) policy for optimal soil moisture irrigation. Another intelligent irrigation system integrates technologies like ESP32, IoT, HTTP, SSE, HTML, and CSS to reduce costs and promote sustainable practices by minimizing drought risks and optimizing water usage [14,9]. These systems offer real-time updates on soil characteristics via web-oriented interfaces.

In livestock management, IoT sensors are employed to monitor disease, stress, feed intake, and humidity, while also automating processes such as milking and feed supply. These systems have been proven to reduce mortality rates by 10 % and increase milk production by 5 %. Examples of IoT implementations also encompass cattle farm management, health monitoring, animal tracking, and improving milk efficiency and productivity [19]. Furthermore, the integration of AI and IoT significantly contributes to soil health, efficient resource management, risk reduction, and stakeholder engagement, thereby enhancing the sustainability of smart agriculture [87]. Wireless sensor networks monitor various parameters and collect data for cloud storage, where WSN/IoT Nodes represent physically installed sensors and WSN/IoT Gateways collect data for transmission to cloud storage for long-term information storage and AI-based analysis. In [70], a cost-effective system for automatic irrigation is presented, which aims to integrate IoT devices with ad-hoc vehicle networks, especially in the authentication process and compatibility features between nodes. Although it is thought to be cost-effective, it has only been tested through simulations and there is no data from practical implementation and real-world practical efficiency. Remote sensing and techniques such as thermal imaging, multispectral imaging, radar remote sensing, satellite

Table 6

A summary of the identified studies and their correlation with the RQs.

Study	Objectives	Relation to RQs
[2]	A blockchain-based framework for tracking the cultivation of agricultural products, giving particular importance to stability and the use of smart contracts	RQ1, RQ2, RQ4
[9]	Intelligent irrigation system using advanced technologies such as ESP32, IoT, HTTP, SSE, HTML and CSS, and the SSE protocol for automatic data updates, the use of DHT22 sensors for accurate and continuous measurement of soil characteristics.	RQ1, RQ2, RQ3
[13]	A framework with a hybrid blockchain structure is proposed which attempts to bring together technologies such as IoT, DL, BT and AI with the aim of solving current problems related to security, data management, energy use.	RQ1, RQ2, RQ4, RQ5
[14]	Advancing efficient and sustainable practices in agriculture, giving importance to cultivation through intelligent greenhouses, which more easily implement digital technologies to increase productivity. Using blockchain and deep learning, traceability in tomato cultivation has been attempted.	RQ1, RQ2, RQ4, RQ5
[19]	IoT applications in smart livestock, IoT network infrastructure for these environments, analysis of many applications for monitoring and tracking animals using IoT, automation of many processes in livestock, analysis of the security of IoT devices in livestock.	RQ1, RQ2, RQ3, RQ5
[24]	Industry 5.0 and the use of remote sensing in agriculture, challenges of integrating remote sensing and applications including data driven analytics, robotics and automation, BT and human-machine interactions.	RQ1, RQ2, RQ4, RQ5
[26]	Blockchain implementation for agriculture supply chain management, Predictive ML models using Python, protection from various attacks using blockchain	RQ1, RQ2, RQ4
[67]	Smart irrigation system for tomato cultivation using Smart irrigation data optimization and robust transmission (SIDORT) Message Queuing Telemetry Transport (MQTT) system	RQ1, RQ2, RQ3
[68]	Secure authentication scheme using hybrid blockchain, with the aim of creating a secure chain to increase the scalability of organizations that are part of the agricultural chain.	RQ1, RQ4
[69]	Blockchain framework for protecting data security and privacy based on Corda DLT, for secure data distribution in IoT ecosystems.	RQ1, RQ2, RQ4
[70]	Integration of Vehicular ad-hoc networks for authentication and communication between intelligent cars and intelligent robots with smart applications in agriculture, security analysis and deep learning techniques for intelligent automatic irrigation	RQ1, RQ2, RQ3, RQ5
[71]	Proposing an architecture including embedded controller, cloud computing and telemetry data in JSON format to increase security in intelligent water management.	RQ1, RQ2
[72]	A scheme is proposed for fire detection through IoT devices on farms using MQTT technology and integrated systems, which enables real-time response and prevention of major disasters.	RQ1, RQ2
[73]	Analysis of blockchain platforms for use in agriculture in combination with IoT devices, specifying Hyperledger Fabric as the most efficient platform in terms of security and privacy.	RQ1, RQ2, RQ4
[74]	Implementing BT for secure data transfer using narrowband IoT is thought to be a solution for addressing data security and efficiency.	RQ1, RQ3, RQ4
[75]	Long-range technologies as an opportunity for efficient transfer in agricultural systems using IoT. The architecture of the LoRa protocol in IoT is studied, the possibility of implementing AI in LoRa.	RQ1, RQ3, RQ5
[76]	Taxonomy for classifying smart agriculture applications in India, also studying network infrastructure, SA sustainability, and implementation possibilities of AI agents towards optimization and rational use of resources.	RQ1, RQ2, RQ3, RQ5
[77]	Monitoring of agricultural systems through IoT and AI, with a special emphasis on service automation, data transfer using long-range technologies without being tied to Global System for Mobile Communications (GSM) technology.	RQ1, RQ2, RQ3, RQ5
[78]	Multi-layered architectures, security and privacy issues, smart farming cyberattacks, supply chain and compliance, trust and information sharing are studied.	RQ1, RQ2, RQ4
[79]	Description of the challenges in adapting blockchain technology, Industry 5.0 and edge computing, creating an innovative architecture that combines BT and edge computing.	RQ1, RQ2, RQ4
[80]	The vision of ML techniques is analyzed to improve efficiency in precision farming, a case study of a computer vision and AI method for detecting objects in depth.	RQ1, RQ2, RQ5
[81]	Models for data collection through IoT, ML and AI techniques for measuring soil characteristics, proposal of a technical architecture for monitoring soil health and automating information collection	RQ1, RQ2, RQ5
[82]	Blockchain-based framework for tracking all IoT-based supply chain users, using smart contracts, a signature scheme is used for identification purposes.	RQ1, RQ2, RQ3, RQ4
[83]	Proposing an architecture based on IoT and BT for efficient water management. IoT devices can interact with BT even over Low Power Wide Area Networks (LPWAN).	RQ1, RQ2, RQ3, RQ4
[84]	Basic concepts of blockchain in agriculture, validation of models using Java programming language, benefits of the proposed system and designed database, results, performance and data security analysis	RQ1, RQ4
[85]	Techniques for increasing security, a framework for preserving data privacy is proposed, based on Privacy-Preserving Blockchain Systems	RQ1, RQ4
[86]	Agri-4-All, a framework based on IoT, blockchain and smart contracts to automate processes within the agricultural supply chain, is presented, and business process models for creating agricultural supply chains are analyzed.	RQ1, RQ2, RQ4
[87]	Applications of digital technologies in the field of agriculture, with particular emphasis on sensors, IoT, WSN, cloud, blockchain. Proposing a blockchain-based platform for monitoring the olive production process, following the cultivation process from the beginning to the final product.	RQ1, RQ2, RQ3, RQ4
[88]	Real-time monitoring and automation, improving agriculture using blockchain and IoT, proposing a software as a service that uses IoT, blockchain, and Vechain for real-time monitoring and weather forecasting.	RQ1, RQ2, RQ4, RQ5
[89]	Wireless application protocols in SA, their architectures, recent IoT-WSN challenges and recommendations for the future development of smart agricultural systems, effectiveness and standardization, AI techniques, emotion analysis for crop analysis	RQ1, RQ2, RQ3, RQ5
[90]	Predictive analytics for efficient resource management in smart agriculture, using the Latent Dirichlet Analysis (LDA) model to predict research areas for smart agriculture	RQ1, RQ2, RQ3
[91]	Application and challenges of edge computing in IoT devices in agriculture, data processing, privacy and security protection, service stability and analysis of future implementations	RQ1, RQ2, RQ4

(continued on next page)

Table 6 (continued)

Study	Objectives	Relation to RQs
[92]	Techniques for the procurement process and customer fulfillment in the agricultural supply chain using BT are studied, case study Parmigiano Reggiano, the combination of BT, Radio Frequency Identification, and the Vendor Managed Inventory (VMI) strategy has shown success	RQ1, RQ2, RQ3, RQ4
[93]	Proposing a model for protection against cyber attacks based on certificates called CBHA and SCKA for smart agriculture, including security mechanisms at each layer of IoT devices.	RQ1, RQ2, RQ4
[94]	Framework for secure communication and authentication between drones and sensors using blockchain technology, namely the SHA-256 algorithm and ECC, as well as using Hyperledger Caliper for proof-of-concept performance examination.	RQ1, RQ3, RQ4

imagery, aerial imagery, UAV are some techniques that are used for crop monitoring, water management, plant disease identification, environmental monitoring, soil health monitoring and others [24].

Despite these implementations, a persistent challenge for the future remains the simultaneous increase in productivity and reduction of costs, especially concerning urban indoor agriculture. While vertical farming is rapidly transforming global agriculture by integrating advanced technologies like AI, robotics, and sustainable practices, simulating artificial conditions to boost productivity in urban areas through indoor agriculture currently presents a cost-efficiency hurdle. This is because it often incurs more expenses than leveraging the natural conditions that fertile soils provide. IoT sensors continuously monitor parameters such as temperature, humidity, and light intensity in vertical farms, providing real-time data for precise management. Despite these advancements, the strong consumer desire for organic products means that these advanced techniques are often applied predominantly in regions experiencing severe droughts or possessing limited fertile soil, where cost becomes a secondary consideration. Nonetheless, IoT-driven adaptive lighting systems show potential for optimizing plant growth in indoor environments, demonstrating scalability and efficiency in precision agriculture.

4.3. The communication protocols between IoT devices and physical agricultural systems (RQ3)

Communication protocols are fundamental to the operation of IoT devices within physical agricultural systems, enabling the flow of data essential for smart farming. MQTT is a common communication method providing sustainable food management by serving as an intermediary among oracles, smart sensors, cloud computing, blockchain, and decentralized databases, facilitating communication between field devices and central servers. Its adoption is often attributed to its efficiency with power-efficient sensors [9]. However, MQTT, like other protocols, faces several challenges, including limited capacity for offline data transfer, inherent security risks, and scalability issues, particularly for larger farms requiring high-capacity data management. Its dependence on intermediaries, such as oracles, also presents limitations [72]. Similarly, WSN and other wireless protocols, while cost-effective and energy-efficient for applications like pest management and moisture monitoring, raise concerns regarding data integrity, privacy, and security [89].

As presented in Fig. 5, the state-of-the-art communication protocols utilized in smart agriculture settings, are broadly categorized into three main groups: long-range (e.g., GPRS, LTE, 3G/4G/5G/6 G), short-range (e.g., Zigbee, WiFi, RFID, Bluetooth), and medium-range (e.g., SigFox, LoRaWAN, NB-IoT). For each category of communication protocols, we have presented the main characteristics, including the main protocols that are part of it, range, power consumption, frequency, and data rate, among others. Long-range

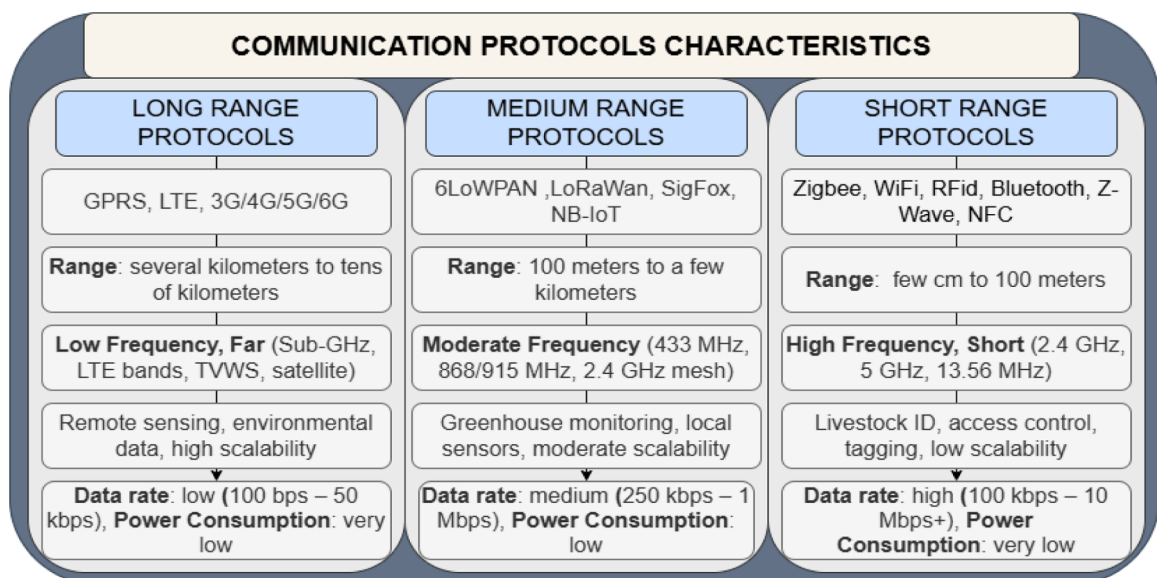


Fig. 5. Classification of the main characteristics of communication protocols in smart agriculture.

protocols, characterized by ranges spanning several to tens of kilometers, are primarily used for communication between edge/fog devices and the cloud, with 5 G and 6 G expected to dominate due to their enhanced security, low latency, and high transmission capabilities. Nevertheless, their deployment in rural areas often faces challenges due to a lack of high-speed backhauls [76]. Medium-range protocols like LoRaWAN and NB-IoT, offering ranges of hundreds of meters to a few kilometers, are typically utilized for communications between end-devices and edge devices. LoRa, distinct from Wi-Fi and Bluetooth, provides longer-range, more efficient, and cost-effective transmission. The integration of AI with LoRaWAN aims to optimize IoT performance in agriculture, enhancing resource allocation and predictive capabilities while addressing challenges such as energy efficiency, interference, and scalability [75]. Short-range protocols, covering distances from a few centimeters to 100 m, are used for various smart device communications and to control intelligent machines. Beyond these classifications, Low-Power Wide Area Networks (LPWAN) are increasingly adopted for longer transmission distances and low energy consumption, offering lower maintenance costs and simpler integration with IoT devices [83].

Vehicle Ad-hoc Networks (VANET) also present a flexible and secure communication method, especially beneficial for irrigation and real-time decision-making in agriculture, with 5 G technology further improving their capabilities [70]. In agriculture, the implementation of VANET architectures would be very efficient, especially for the irrigation of fertile lands in different areas, uninterrupted communication between different sensors and above all predictive analysis and real-time decision-making for the execution of various processes related to agricultural processes.

Security vulnerabilities pose significant threats across various layers of IoT agricultural systems. Data transfer is particularly susceptible to attacks at the communication layer. As shown in Fig. 6, prevailing threats include routing attacks, Denial-of-Service (DoS) attacks targeting servers and disrupting system functions, data transmission attacks, malware injection, spoofing and impersonation, and radio frequency jamming. Ensuring confidentiality, integrity, data update, authentication, and non-repudiation is crucial for each IoT sensor [78].

Mitigation strategies for these security vulnerabilities are multifaceted. Blockchain technology, characterized by address space, data integrity, authentication, authorization, privacy, traceability, and transparency, is considered a superior alternative for data security and protection [73]. Public key cryptography, such as Elliptic-curve Public-Key Cryptography (ECC), is widely used in blockchain for secure authentication and transaction validation. While Ethereum offers openness and decentralization, platforms like Hyperledger Fabric utilize a Public Key Infrastructure (PKI) to verify participant identities, enhancing the security and authentication of transactions and communications. A blockchain framework for safe communication among intelligent equipment and sensors utilizing the Ethereum platform, private blockchain, and smart contracts is described in [94]. The ECC and SHA-256 algorithms are used for safe authentication, designated for asymmetric key exchange via smart contracts. The data is kept in IPFS (InterPlanetary File System), where sensors and intelligent devices concurrently share information processed via the blockchain network. The proof of concept has served as a developmental phase to facilitate authentication and connectivity between the intelligent machine and its associated sensors; however, several shortcomings persist in this area, particularly regarding scalability, transaction latency, execution duration, and limited capacity, among others.

General security improvements include implementing regular password updates and multi-factor authentication, as well as utilizing secure communication protocols and Virtual Private Networks (VPNs) for encrypted communication between IoT devices, systems, and cloud platforms [95]. Overall, improving agricultural IoT security necessitates identifying key threats and raising awareness.



Fig. 6. Security threats of IoT-enabled agricultural systems.

4.4. Achieving security, data integration and automation through IoT, AI, and blockchain (RQ4)

The seamless integration of IoT, AI, and blockchain technology in agriculture presents significant challenges related to security, privacy, and data integration, while also aiming for enhanced service automation without third-party involvement]. Even though wireless devices offer flexibility, they inherently increase the risk of various attacks on data. Consequently, there is a growing need for robust frameworks and solutions to address these vulnerabilities and ensure secure, reliable, and automated agricultural processes. A notable trend in communication protocols is the adoption of Low-Power Wide Area Networks (LPWANs), including NB-IoT, LoRa, and SigFox, which are favored for their low electricity consumption and wider communication wave distribution [83,85]. These technologies offer longer transmission distances, lower maintenance costs, and simpler integration with IoT devices, supporting the widespread deployment of sensors in smart agriculture. However, concerns about data privacy and security are escalating due to the growing volume of real-time data processing in smart agriculture. Agriculture 5.0 is envisioned to address these issues, focusing on advanced resource management and decision-making processes facilitated by secure data handling [77]. Proposed security measures encompass stringent data access controls, encryption, data minimization, anonymization, and real-time monitoring. Furthermore, processing data at the fog and edge layers, along with developing scalable and reliable infrastructures, are crucial for bolstering the security of smart agriculture systems. These measures are vital for preventing data manipulation and various cyber attacks [96].

Blockchain technology plays a critical role when synergistically combined with IoT and AI, offering significant benefits in data protection, privacy preservation, and integrity. BT can be utilized for secure storage and big data analytics of information collected by IoT devices. While much research on BT in agriculture remains conceptual, some studies demonstrate its successful application in water management and agricultural rights, leading to cost reductions and increased irrigation efficiency [67,70]. Frameworks like the Hybrid blockchain-based many-to-many cross-domain authentication (HBMA) scheme are designed to enhance security in smart agriculture IoT networks through decentralization, which ensures network resilience even if some nodes are compromised, and mutual authentication, which provides a secure verification process for transactions [97]. Additionally, blockchain ensures privacy preservation through autonomous authentication protocols, perfect forward secrecy for key exchange, and resistance to Distributed Denial of Service (DDoS) attacks. This technology also aids in fraud detection, fostering trust among participants, and enabling transparent product traceability throughout the agricultural chain [69,24]. The integration of BT into IoT can occur either by having IoT devices act as blockchain nodes or by treating BT as a service integrated into IoT architectures [78,93]. Smart contracts are pivotal in automating various agricultural processes, including transaction processing, supply chain management, and compliance monitoring, fostering equality and mitigating fraud. The standardization of protocols and operations is crucial for the effective execution of smart contracts and seamless interaction between BT and IoT devices [80]. Furthermore, the combination of IoT and AI significantly contributes to soil health, enables predictive analytics through techniques like digital twin simulations, and enhances decision-making for optimal agricultural interventions and automation [81].

The synergistic combination of blockchain technology, the IoT, and AI is instrumental in enabling digital transformation and advanced agri-data management. IoT, which involves devices with sensors and processing capabilities that exchange data, facilitates real-time processing and monitoring. When integrated with AI and machine learning, it empowers the use of intelligent algorithms, including those deployed in UAVs and autonomous AI agents. This integration enhances service quality, provides sophisticated data analysis, and improves decision-making, thereby increasing productivity and reducing both costs and operational time. Furthermore,

Table 7
Socio-technical barriers.

Barrier (social vs. technical)	Type of barrier	IoT impact	AI impact	Blockchain impact
Social	High cost	Expensive sensor/device deployment	High costs for data processing and model training	Costly infrastructure setup
	Digital divide and skills gap	Limited digital literacy for device use	Shortage of skills for model development and interpretation	Lack of expertise for secure management
	Loss of local knowledge and labor displacement	Automation reduces traditional roles	Risk of replacing human expertise	Less direct impact
	Resistance to change and trust issues	Skepticism about device reliability	Reluctance to rely on automated decisions	Distrust in decentralized systems
	Regulatory and governance uncertainty	Unclear device/data regulations	Uncertain rules for data use and automation	Legal ambiguity for data ownership
Technical	Data security and privacy concerns	Vulnerable to breaches and manipulation	Sensitive data exposure in analytics	Complex privacy management, immutable data
	Integration complexity	Difficult to connect heterogeneous devices	Complex to integrate with real-time data streams	Challenging to link with legacy systems
	Scalability and interoperability issues	Difficult to scale sensor networks	Challenges in handling large, diverse datasets	Throughput and latency limitations
	Lack of standardized frameworks	No common protocols for device communication	Difficulty in sharing and comparing models/data	Inconsistent data and compliance standards
	Poor data quality	Sensor errors and noise	Garbage-in, garbage-out in analytics	Immutable storage of bad data
	High computational requirements	Limited by edge device power	Resource-heavy model training and inference	Energy-intensive consensus protocols

BT significantly contributes to data protection, ensuring privacy preservation, integrity, and the detection of potential manipulations and various cyber-attacks, especially when combined with AI agents [14,90,94].

Addressing data integration and interoperability is a critical challenge to ensure seamless data flow across heterogeneous IoT, AI, and blockchain systems. Frameworks that involve standardizing data formats and communication protocols to improve efficiency, which integrate IoT, blockchain, and AI for a scalable and secure agricultural framework have been proposed [89]. Another unified framework leveraging cryptography, blockchain, AI, and IoT for digital ecosystem security that guarantees safe, fast, seamless, and transparent data processing in precision agriculture is suggested by [13]. Nevertheless, the absence of robust standardization and interoperability presents a significant obstacle to the widespread automation of agricultural services, especially within smart irrigation systems [19]. The fragmented agricultural IoT market, characterized by different manufacturers employing proprietary protocols and data formats, complicates the seamless integration of diverse components from various vendors into a unified system. This fragmentation not only leads to elevated operational costs but also limits the scope and sophistication of deployable services. Furthermore, integrating existing or conventional infrastructure with advanced smart agriculture technologies often proves to be a complex and challenging endeavor. Despite these difficulties, an effective approach involves reusing existing components and strategically integrating IoT devices, as demonstrated by practical implementations in irrigation systems that successfully blend traditional practices with modern solutions [76]. However, more research efforts are needed to be actively focused on developing standardized protocols to establish an interoperable environment for smart agriculture, facilitating secure data exchange between disparate precision agriculture systems and devices.

4.5. Socio-technical barriers to full deployment of smart agriculture through the integration of IoT, AI and blockchain - (RQ5)

The full deployment of smart agriculture using IoT, BT, and AI offers significant potential, but it is often hindered by a set of socio-technical barriers. These barriers arise from the interaction between social systems (people, organizations, policies/regulations) and technical systems (infrastructure, tools, algorithms, data). These barriers are interconnected and impact each technology in unique ways. Table 7 summarizes how each barrier interacts with IoT, AI, and blockchain in the context of smart agriculture deployment. Barriers are divided into technical and social, their types are categorized, and the impacts that the implementation of IoT, AI and BT have on each barrier type in particular are described.

4.5.1. Social barriers

One of the most prominent challenges is the *high initial cost* associated with deploying these advanced technologies. The installation and maintenance of IoT sensors and devices, the establishment of blockchain infrastructure, and the development of sophisticated AI models all require substantial financial investment. For many smallholder farmers and organizations operating in resource-constrained environments, these expenses can be prohibitive, limiting widespread adoption. The shortage of expertise is symptomatic of a broader *digital divide and skills gap*, particularly acute in rural areas. Many agricultural communities lack access to the necessary technology and digital literacy required to manage and interpret data from these systems. This gap not only restricts the use of smart agriculture solutions but also exacerbates existing inequalities [69].

Another significant barrier is the risk of *loss of local knowledge and labor displacement*. As smart agriculture technologies become more prevalent, there is a tendency to prioritize automated and data-driven solutions over traditional farming practices and experiential knowledge. This shift can marginalize the valuable insights that farmers have developed over generations, which are often crucial for adapting to local environmental and climatic conditions. Research shows that local knowledge enables farmers to creatively adapt technologies to their specific contexts, combining traditional methods with modern tools to achieve practical and resilient solutions [14]. When technology fails or proves unsuitable, farmers frequently revert to these established practices, underscoring their enduring relevance. However, if the adoption of IoT, blockchain, and AI leads to the erosion of this knowledge base, it could reduce the adaptability and sustainability of agricultural systems, particularly for smallholders and in diverse agro-ecological settings. Moreover, the automation of agricultural processes can result in labor displacement, as tasks once performed by humans are increasingly managed by machines and algorithms. This raises social and ethical concerns, especially in rural communities where agriculture is a primary source of employment and social cohesion. *Resistance to change* and skepticism towards automated and decentralized systems are common among farmers and agricultural stakeholders. *Trust issues* can arise, especially when traditional knowledge and practices are perceived to be undervalued or replaced by technology [73,72].

In addition to these challenges, *regulatory and governance uncertainty* presents a formidable barrier to the integration of advanced digital technologies in agriculture [79,91]. The legal frameworks governing data ownership, privacy, sharing, and cross-border transactions are often unclear or inconsistent, especially with the introduction of blockchain-based record-keeping and AI-driven decision-making. This uncertainty can create hesitation among stakeholders, slow down innovation, and complicate compliance for both technology providers and end users. Without robust and harmonized regulations, there is a risk that smart agriculture systems may not be implemented in ways that are transparent, fair, or aligned with broader societal goals. Addressing these governance issues is essential to building trust, ensuring accountability, and enabling the responsible scaling of smart agriculture initiatives.

4.5.2. Technical barriers

Among the leading technical barriers are persistent concerns over *data privacy and security*. These concerns are amplified by the interconnected nature of these technologies [87]. The aggregation and transmission of sensitive agricultural and personal data across IoT networks, its storage on immutable blockchain records, and its processing by AI systems all introduce vulnerabilities to breaches and misuse. Managing privacy in such a complex ecosystem is a daunting task, especially when regulatory frameworks are still

evolving. *Integration complexity* further complicates the landscape. The need to connect a diverse array of devices, platforms, and protocols - often across legacy and modern systems, presents significant technical hurdles [89]. Achieving seamless data flow between IoT sensors, blockchain ledgers, and AI analytics demands specialized expertise, which is often in short supply.

Scalability and interoperability issues also pose major obstacles [80]. As the number of connected devices and the volume of data increase, ensuring that systems can scale efficiently while maintaining seamless data exchange becomes increasingly challenging. Blockchain and AI systems, in particular, face throughput and latency limitations that can hinder real-time agricultural operations. The *lack of standardized frameworks* for data, protocols, and compliance further complicates integration and regulatory approval. Without common standards, interoperability between different devices and platforms becomes challenging, leading to fragmented systems and inefficiencies. Furthermore, the lack of clear guidelines complicates regulatory approval and can discourage both investment and innovation in the sector. *Data quality* is another technical barrier that cannot be overlooked [69,88]. Sensor errors and environmental noise can lead to the collection and storage of unreliable data, which, when recorded immutably on blockchain or used in AI analytics, can undermine the effectiveness of smart agriculture solutions.

Finally, the *high computational requirements* of AI models and blockchain operations present practical challenges, particularly in rural or resource-limited settings where access to reliable power and processing infrastructure may be lacking [24].

4.6. Benefits of integrating IoT, AI, and blockchain in agriculture

Integrating IoT, BT and AI in agriculture enables a leap in efficiency, precision, and security. Automation technologies facilitate data-driven decisions for irrigation, fertilization, pest control, and harvest timing, all of which enhance productivity. Blockchain contributes to transparency and traceability in agricultural supply chains, ensuring secure transactions and fostering trust among stakeholders. Advanced intelligent algorithms used in the literature such as Linked Least Traversal (LLT) and Optimal Soil Wetness Closeness Policy (OSWCP) illustrate how integrated systems can optimize resource use. These models support precision irrigation and multi-layer soil nutrient monitoring by leveraging IoT sensors and AI analytics. This not only reduces water and fertilizer waste but also boosts crop yield, particularly for high demand crops like tomatoes [67].

Emerging technologies aligned with Agriculture 5.0, such as 6 G networks, quantum computing, and digital twins, expand the possibilities for connected and autonomous farming systems. Combined with IoT, AI, and BT, these advances enable real-time monitoring, improved crop modeling, disaster prediction, and remote collaboration through robotic systems, even in isolated rural areas [37,38,40,65,66]. Edge and fog computing also provide meaningful benefits in real-time data processing for smart agriculture. These technologies reduce latency and network load by handling data close to its source. When integrated with AI, they allow for timely decision-making in applications like automated irrigation and disease detection, significantly improving reliability and scalability in large-scale deployments [68,93].

AI-enabled UAVs and drones enhance field surveillance by capturing detailed imagery for crop health, soil moisture, and livestock monitoring. These visual insights help in identifying issues earlier and implementing targeted interventions. Automated systems reduce labor costs, minimize errors, and allow continuous operation, thereby improving overall farm productivity and operational sustainability [83,94]. Beyond production-level benefits, Blockchain supports secure data exchange and provenance tracking across complex agricultural value chains. This is crucial in organic and fair-trade farming, where transparency about the origin, handling, and quality of produce is essential. Smart contracts also reduce administrative burdens and automate transactions, enhancing speed and accuracy in farm operations [69,73].

In Fig. 7, we present a holistic architecture demonstrating how the integration of IoT, AI, and blockchain technologies can

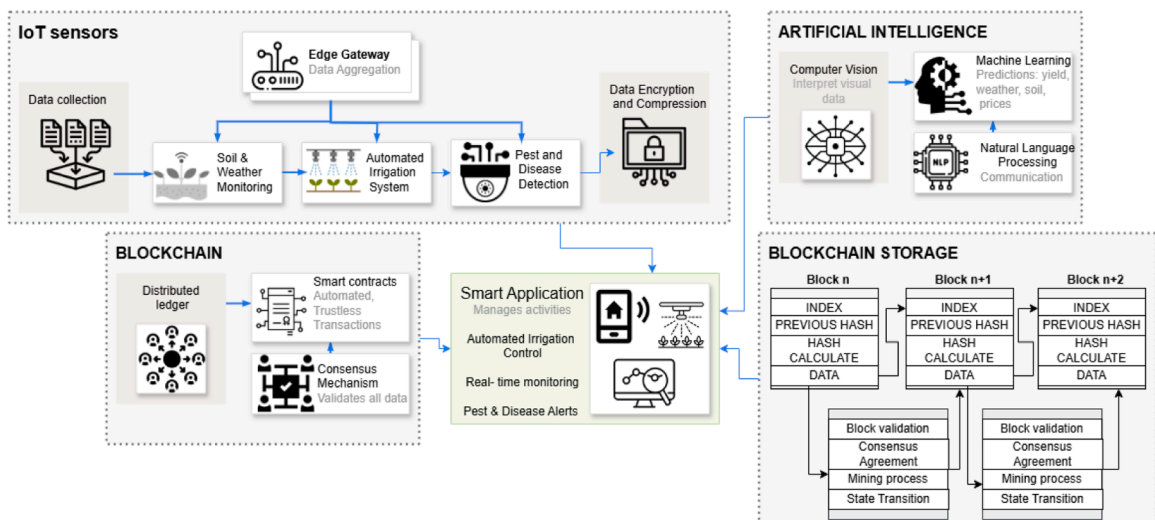


Fig. 7. A holistic architecture for integrating AI and BT into IoT devices for smart agriculture.

revolutionize modern farming practices through intelligent automation, secure data handling, and advanced analytics.

Next, we will provide a detailed description of the proposed architecture through Fig. 7. The process begins with IoT sensors distributed throughout the agricultural environment, continuously collecting data on soil conditions, weather, crop health, and pest presence. This raw data is aggregated via an edge gateway, which serves as a local hub for processing and managing the information before it is transmitted further. The aggregated data supports critical applications such as soil and weather monitoring, automated irrigation systems, and pest and disease detection, ensuring that the farm operates with optimal resource efficiency and timely responses to environmental changes.

Before this sensitive information is shared with other parts of the system, it undergoes data encryption and compression to ensure secure and efficient transmission. Once secured, the data is leveraged by AI modules. Here, computer vision algorithms interpret visual data from the field, such as identifying crop diseases or assessing growth stages, while machine learning models generate predictions related to yield, weather patterns, soil health, and even market prices. Natural language processing further enhances the system by enabling intuitive communication and reporting, making insights accessible to farmers and stakeholders. Simultaneously, the data is integrated into a blockchain infrastructure. The blockchain acts as a distributed ledger, recording all transactions and data entries in a transparent and immutable manner. Smart contracts automate and enforce trustless transactions, such as triggering irrigation based on sensor data or processing payments for produce. The consensus mechanism within the blockchain validates all data entries, ensuring that only accurate and agreed-upon information is stored.

A central smart application orchestrates these activities, managing automated irrigation control, real-time monitoring, and generating alerts for pest and disease outbreaks. This application serves as the user interface, connecting seamlessly with mobile devices, drones, and other smart equipment to provide actionable insights and enable remote management of farm operations. Finally, all validated data and transactions are stored in the blockchain storage system. Each block contains an index, a reference to the previous block (hash), a newly calculated hash, and the data itself. The process of block validation, consensus agreement, and mining ensures that every piece of information is securely recorded and that the system's state transitions are transparent and auditable. Through these tightly integrated components, the architecture enables a new era of smart agriculture, one that is automated, data-driven, secure, and sustainable, embodying the principles of Agriculture 5.0. Lastly, it is worth mentioning that these technologies pave the way for climate-smart agriculture. AI can model climate risks and simulate adaptation strategies, while IoT devices continuously monitor environmental parameters. Together, these systems inform sustainable practices, reduce environmental impact, and help farmers adapt to shifting weather patterns, ensuring long-term resilience and food security [78,24].

4.7. Implications and recommendations

The integration of IoT, blockchain, and AI into smart agriculture has opened new pathways for efficiency and automation, yet its practical implications are still underexplored in existing research. Many studies included in this systematic review paper offer limited empirical insights or practical frameworks for addressing real-world agricultural challenges. However, key implications identified relate to *traceability, transparency, secure certification processes, fraud detection, and compliance with agricultural standards*, as discussed in [2]. In terms of operational application, intelligent irrigation systems show potential when adapted to local environmental and agricultural contexts. These systems facilitate informed decision-making through real-time feedback, especially when integrated with intelligent machinery [71]. Nevertheless, their deployment reveals important concerns around *cost-effectiveness, equipment durability, data security, and user training*, which are recurring themes in the literature [72].

Other implications relate to the *technical complexity and hardware constraints* of smart systems. These include data overload, limited battery life, range and coverage limitations, and the need for accurate sensor calibration. Moreover, data transmission stability and the ability to prevent interference, especially in edge computing environments, pose significant technical barriers that limit scalability and reliability [87–89]. Crop yield prediction, which depends on a mix of soil quality, atmospheric conditions, and other environmental variables, presents another layer of challenge for AI systems. Accurate predictions demand diverse, high-quality datasets and continuous learning mechanisms to adapt to fluctuating field conditions, an issue underscored in [81]. These implications show that while automation holds promise, *context-aware systems* are essential for long-term success.

Communication protocols used in agriculture, such as LoRaWAN for long-range low-power connectivity, 5 G for low latency, Wi-Fi for bandwidth, and Bluetooth for energy efficiency, each have specific roles and trade-offs. The implication is that a *hybrid communication model*, tailored to geographic conditions and user needs, is necessary to support robust smart agriculture infrastructure [83]. Blockchain offers unique strengths in *identity management, secure authentication, and decentralized data governance*, making it critical for combatting data tampering and cyberattacks in agricultural environments. However, it remains vulnerable if the data sources themselves (e.g., sensors) are inaccurate or compromised. Hence, the reliability of *physical IoT components* remains central to blockchain's effectiveness in this context [69,77–78]. Furthermore, smart contracts provide valuable automation benefits, such as facilitating *online payments, supply chain tokenization, and autonomous marketplaces*. Yet their execution in agriculture is hindered by challenges such as high costs, limited interoperability, infrastructure gaps, and execution bugs. These limitations reflect broader concerns about the *practical deployment of AI and BT* for real-time, autonomous decision-making in unpredictable farming environments [87]. Ultimately, the *farmer remains a vital part of the agricultural intelligence system*, not only as a system operator but also as a creative problem-solver in the face of unexpected situations. Future endeavors should focus on the *co-evolution of human and machine intelligence*, where AI systems continuously learn from farmers' decisions and behavior to improve adaptability, reliability, and user trust [81].

Based on the findings of this study, we highlight the following research ideas and recommendations that researchers, policymakers, technology developers, and practitioners can explore when working on seamless integration of IoT, blockchain and AI in smart

agriculture settings.

Recommendations for researchers should be focused in system design, theoretical modeling, technical feasibility and adaptive frameworks, as emphasized below:

- *Develop scalable and lightweight blockchain consensus mechanisms* to improve transaction speed, reduce computational load, and enable seamless on-chain and off-chain agricultural integrations [73].
- *Encourage modular AI architectures* that support semi-autonomous decision-making, allowing human oversight in critical agricultural operations [81].
- *Support context-aware irrigation systems* using predictive models that consider soil type, crop needs, and environmental factors to optimize resource use [74].
- *Integrate AI models with human-in-the-loop feedback*, ensuring that automated systems can learn from farmer behavior and adapt to unforeseen conditions [88].

A recent pilot study applied an AI-blockchain-based traceability model in the agricultural supply chain to improve transparency, traceability, and efficiency, addressing supply chain fragmentation and enabling secure and efficient data transaction [98].

Recommendations for policymakers emphasize regulation, governance, funding, infrastructure planning, and data rights, as highlighted below:

- *Encourage multi-stakeholder collaboration* among farmers, technologists, scientists, and public agencies to address standardization, ethical guidelines, and interoperability challenge [89].
- *Mandate privacy-by-design practices* in all smart agriculture systems, with regulations focusing on secure identity management, encrypted communication, and transparency in data ownership and use [78].
- *Promote hybrid communication infrastructures* through policy and funding support, enabling adoption of multi-protocol systems like LoRaWAN, 5 G, and NB-IoT in underserved rural areas [83].

The usefulness of policy-backed infrastructure design is shown by a recent research that compares real-world deployments of low-power wide-area networks (LPWAN), fifth-generation (5 G), and hybrid models. The study reveals cost savings of up to thirty percent and enhanced dependability [99].

When it comes to technology developers, they should focus on design, engineering, deployment and maintenance of IoT/AI/BT systems, more specifically:

- *Enhance sensor calibration and reliability* to improve quality of raw data feeding into AI decision systems and blockchain records [77].
- *Create backup and failover mechanisms* for smart agriculture networks to ensure data continuity and system resilience during outages or hardware failures [87].
- *Design user-friendly training tools and interfaces* that simplify smart agriculture operations for non-technical users while maintaining system security and control [72].

A new technical solution developed in Spain describes a technique that uses machine learning to build high-resolution maps of irrigated olive trees. This approach enables improved water management and demonstrates how data-driven decision-making may be operationalized in agricultural contexts [21].

Lastly, recommendations for practitioners are focused on practical application, daily operations, and local adaptation of technologies, as following:

- *Develop cost-effective training programs* focused on practical skills in using IoT devices, managing privacy settings, and responding to system faults [78].
- *Implement real-time monitoring tools* that provide actionable insights and suggestions, while allowing farmers to adjust or override automated decisions when necessary [81].
- *Adopt a flexible communication model* by selecting appropriate protocols (e.g., Wi-Fi, NB-IoT) based on farm size, geography, and connectivity needs to ensure reliable data transfer [83].

For instance, one practical solution which represents an AI-based responsive monitoring algorithm, referred to as the Dynamic-Adaptive Search (D-AS) algorithm that optimizes crop stress management is recently developed. The system operates in real time, dynamically adapting based on immediate input data, improving responsiveness, cost-effectiveness, and resilience. Importantly, it enables human oversight of automated decisions, aligning well with farmer autonomy and system reliability needs [100].

We believe that the successful integration of IoT, AI, and blockchain in agriculture depends not only on technological innovation but also on coordinated action across research, policy, industry, and practice. Each stakeholder, including researchers, policymakers, technology developers, and practitioners, has a critical role to play in ensuring that these technologies are accessible, secure, adaptive, and scalable. By addressing the identified challenges through the targeted recommendations above, the agricultural sector can move toward a more sustainable, automated, and resilient future, where intelligent systems support productivity without compromising data privacy, ethical integrity, or farmer autonomy.

5. Conclusion

This systematic literature review has examined the integration of IoT, blockchain technology, and AI within the domain of smart agriculture, emphasizing their combined potential to enhance productivity, efficiency, security, and automation in agricultural practices. By analyzing 35 high-quality studies published between 2019 and 2025, we identified key benefits such as real-time data collection and monitoring through IoT, secure and transparent supply chains via blockchain, and intelligent decision-making supported by AI-driven analytics.

Despite their individual and collective promise, the review revealed several persistent challenges and gaps, such as data privacy and security, limited infrastructure in rural areas, lack of interoperability between systems, high implementation costs, and user resistance due to low digital literacy. Furthermore, the practical deployment of these technologies faces technical hurdles in terms of energy efficiency, network reliability, and sensor accuracy, as well as socio-ethical concerns related to data ownership and autonomy in automated decision-making. The findings underscore the importance of a multi-disciplinary approach, involving researchers, policymakers, technology developers, and farmers to co-design context-aware, scalable, and user-centric solutions. There is also a clear need for stronger policy frameworks, hybrid communication models, improved training programs, and the development of adaptive AI systems that can incorporate human feedback. Ultimately, although the integration of IoT, blockchain technology, and AI holds transformative potential for smart agriculture, realizing this vision requires addressing technical and social barriers in parallel. The insights and recommendations presented in this review can serve as a foundation for future research, development, and policy-making aimed at building secure, automated, and sustainable agricultural systems.

CRediT authorship contribution statement

Avni Rustemi: Writing – original draft, Visualization, Project administration, Investigation, Data curation, Conceptualization.
Fisnik Dalipi: Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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